

1. L1 loss is  $|x-y|$     L2 loss is  $(x-y)^2$

Gradient for L1  $\frac{\partial L_1}{\partial x} = \text{Sign}(x-y)$

this is always 1, -1 or 0

when  $x > y$ ,  $x < y$  or  $x = y$

So the updated  $X = X_{\text{old}} - \underset{\text{learning rate}}{\alpha} \cdot \underset{\text{old}}{\text{Sign}(x-y)}$

This is linear, so it can have abrupt changes since it normally is +1 or -1, it can deviate a lot

if  $x_i$  is close to  $y_i$ .

Gradient for L2

L2 loss is squaring the difference between predicted & actual,

$$\frac{1}{2} \sum (x_i - y_i)^2$$

This tells us how to change  $x$  to be as close to  $y$  as possible. Since it's squared, it is able to have a smooth approach compared to a abrupt change like L1.

In summary, L2 is smooth so  $x$  gets more smaller & precise as it approach the desired value, But L1 still changes the same as we ~~can~~ go closer to desired, so it takes longer to finally approach it.

$$2. \quad L(\hat{y}, y) = - \sum_{i=1}^n y_i \log(\hat{y}_i)$$

Since  $y = [1, 0, \dots, 0]$

$$\text{So } L(\hat{y}, y) = -\log(\hat{y}_1)$$

Since we only need to care for the first term

$$i. \quad L(\hat{y}, y) = 0$$

$$-\log(\hat{y}_1) = 0$$

$$\log(\hat{y}_1) = 0 \quad \hat{y}_1 = 1$$

$$\text{So } \hat{y} = [1, 0, 0, \dots, 0]$$

$$ii \quad L(\hat{y}, y) = \infty$$

$$-\log(\hat{y}_1) = \infty$$

$$\log(\hat{y}_1) = -\infty$$

$$\hat{y}_1 = 0$$

$$\hat{y} = [0, \hat{y}_2, \hat{y}_3, \dots, \hat{y}_c]$$

These are non-negative and they sum up to 1, since they are probabilities